Adaptive Segmentation and Automated Morphometric Analysis of Monogenetic Volcanic Cones in Distributed Fields

Aubrey Schrameck[1] and J-F Smekens[1]

[1][Northern Arizona University, Flagstaff, Arizona]

Background and Motivation

Distributed volcanic fields preserve both primary eruptive morphology and post-eruptive geomorphic modification over thousands to millions of years. High-resolution DEMs now enable quantitative morphometric analysis at regional to planetary scales.

While morphometric parameters can provide meaningful distinctions among volcanic landforms, several limitations persist:

- Morphometric variables often overlap across eruptive styles, reducing diagnostic power (Pedrazzi et al., 2024).
- Post-eruptive erosion systematically modifies cone geometry (McGuire et al., 2014).
- Small sample sizes and inconsistent segmentation workflows limit reproducibility
- Extremely large vent populations (e.g., >65,000 mapped on Mars) require automated, scalable methods.

Core Goals

A reproducible, scalable, and adaptive pipeline that:

- Standardizes DEM segmentation
- Extracts both primary and asymmetry metrics
- Enables multivariate statistics and machine-learning analyses
- Scales from regional Earth datasets to planetary vent populations
- Incorporates automated error handling, persistent logs, and diagnostic warnings

Area of Study

- Applied to 445 cones in the San Francisco Volcanic Field (Northern Arizona)
- Used 1-m DEMs from USGS 3D Elevation Program

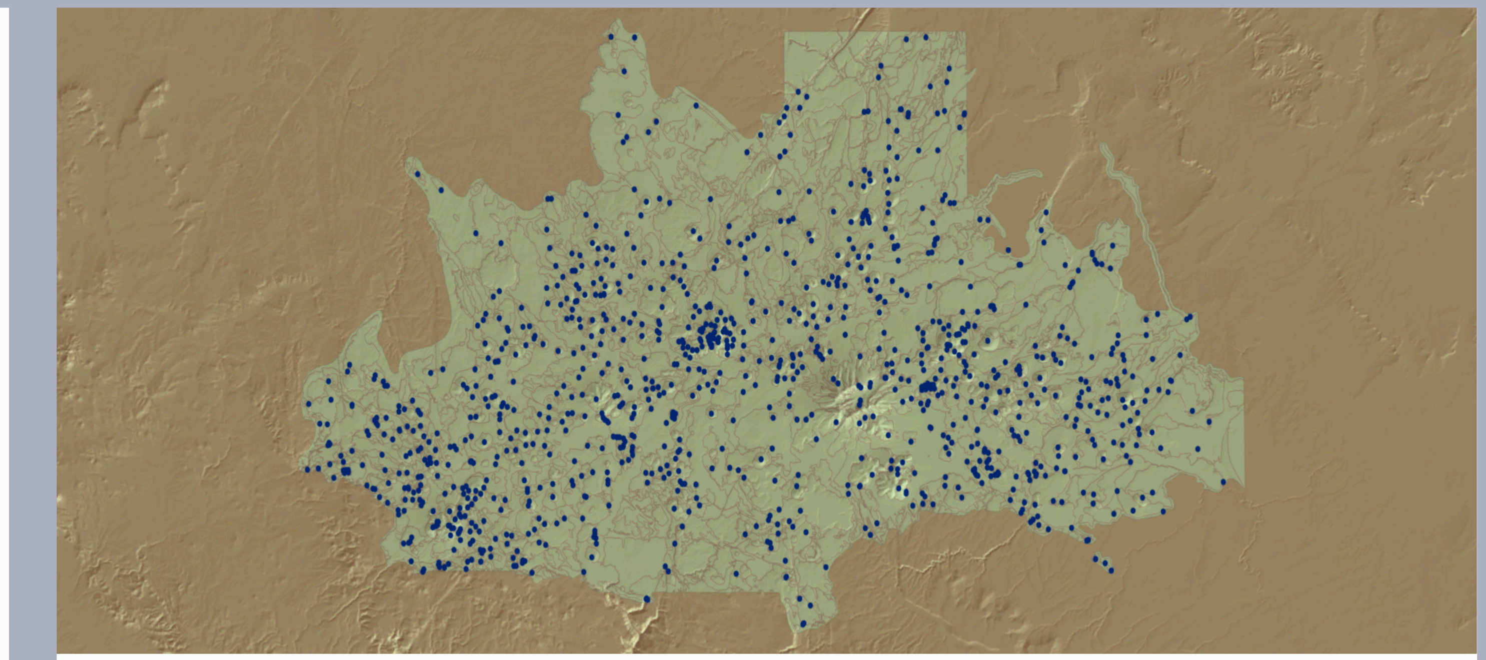


Figure 1: San Francisco Volcanic Field vent coordinates

Stage 1: Adaptive DEM Segmentation

Objective

Automatically retrieve, validate, and segment 1-m DEM data to delineate crater rims and cone bases using adaptive radial sampling with built-in quality control.

Inputs: vent number, center coordinates (lat, lon), output directories (DEM + polygons), and optional diagnostic flag.

Dynamic DEM Retrieval

- Iterative search radius expansion (1 km increments)
- Query USGS 3D Elevation Program via TNM Access API
- Download intersecting DEM tiles
- Tile integrity validation (window read + corruption check)
- Mosaic validated tiles into a single GeoTIFF

Radial Sampling Algorithm

72 radials at 5° increments, elevation sampled every 0.5 m along each radial. Search radius expands until a stable edge is detected or the maximum radius is reached.

Crater Rim Detection

- Identify the local elevation maximum within the rim search radius
- Flat-terrain pre-check prevents false positives

Cone Base Detection

Base determined using multiple criteria:

- Minimum elevation beyond the rim
- Sustained rise or flat (≥ 3 points)
- Slope inflection exceeds threshold
- Fallback to search radius if needed

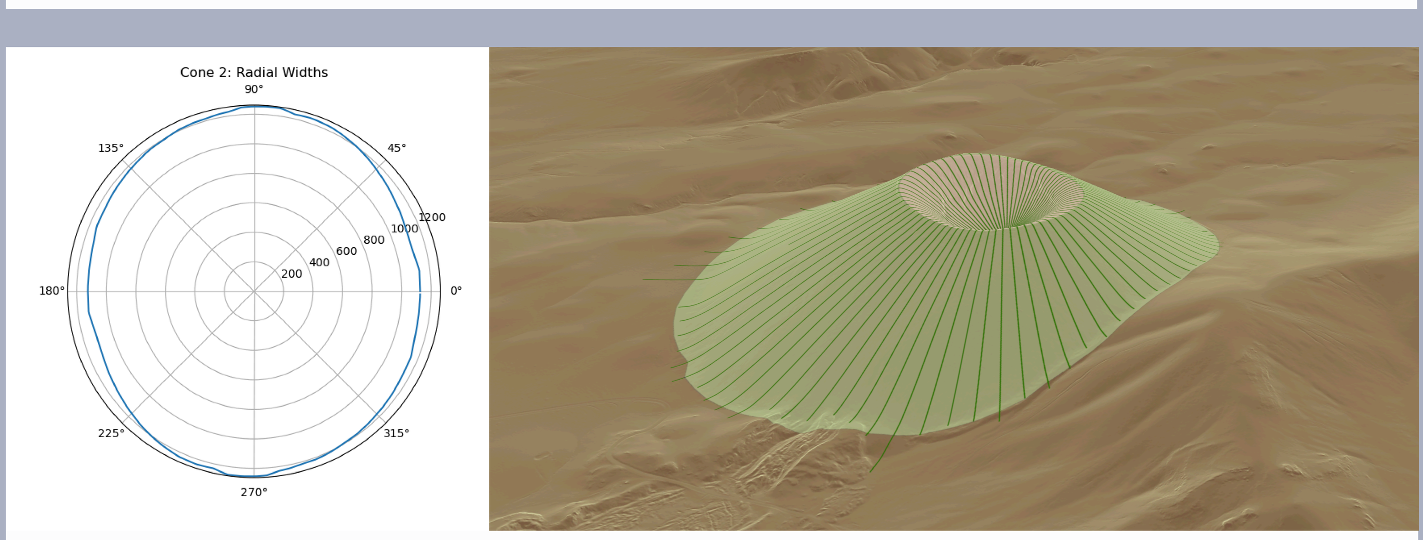


Figure 2: Cone 2 (right) and a polar graph of its radial widths (left), located at 35° 34' 56.49", -111° 37' 56.11"

Polygon Creation & Smoothing

- Moving-window smoothing applied to radial distances
- Smoothed endpoints interpolated into cone and crater polygons.
- DEM clipped to cone polygon

Outputs: Segmented DEM (GeoTIFF), cone and crater polygons (shapefiles), warning flags, and optional diagnostic messages and radial lines.

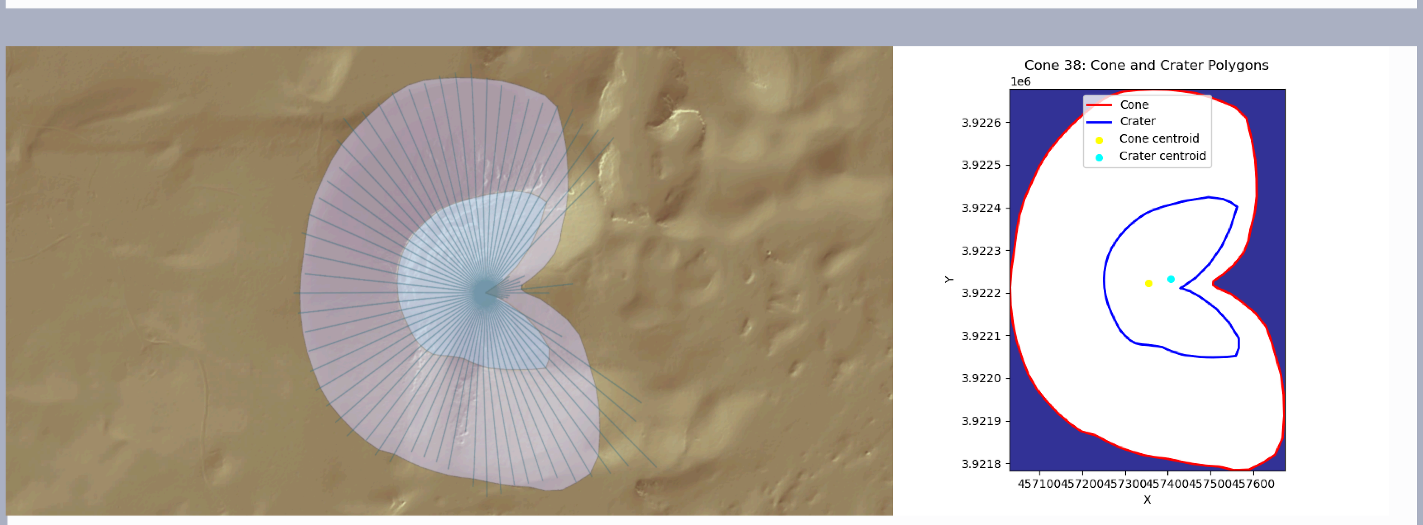


Figure 3: Cone 38 (left) and its associated polygons (right), located at 35° 26' 32.77", -111° 28' 8.54"

Stage 2: Morphometric Extraction

Objective

Quantitatively extract morphometrics from segmented cone and crater polygons, including basal-surface–corrected height and volume metrics.

Inputs: Segmented DEM, cone and crater polygons, warning reasons, and optional diagnostic flag.

Metrics

General statistics: min, max, mean, median, standard deviation, skew, and kurtosis.

Cone

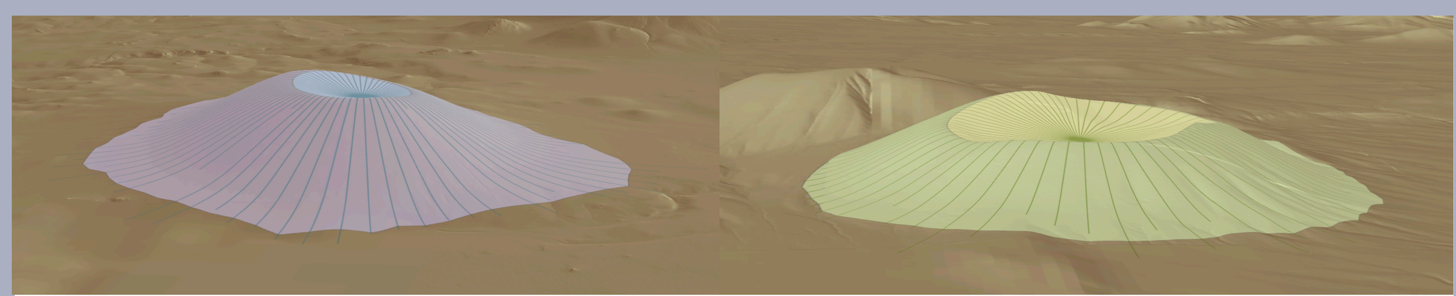
- Max Height
- Volume
- Perimeter/Area
- Elevation statistics
- Width Statistics
- Slope Statistics

Crater

- Max depth
- Fill Volume
- Perimeter/Area
- Width statistics
- Slope statistics

Calculated

- Elongation
- Circularity
- Eccentricity
- Major/Minor Axis
- Orientation
- Ratios



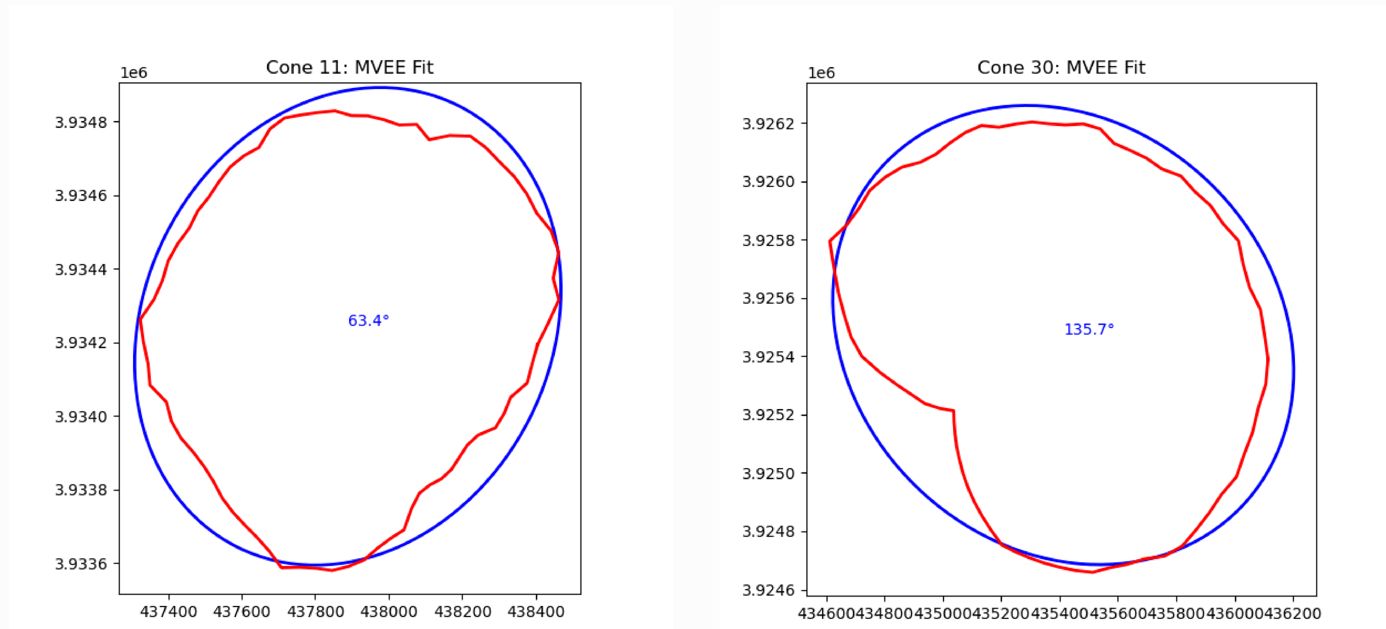


Figure 4: MVEE fit plots for Cone 11 (left), located at 35° 32' 58.12", -111° 41' 6.73", and Cone 30 (right), located at 35° 28' 11.86", -111° 42' 47.2". Red indicates the cone boundary, and blue represents the fitted ellipse

Basal Surface Interpolation

To remove pre-eruptive topographic bias, a basal surface is reconstructed beneath each cone:

- Ring sampling exterior to cone footprint
- Rasterized mask extraction
- Planar (order=1) least-squares surface fit
- Evaluated across the full DEM grid

Height-above-basal DEM: Relief=DEM−Basal Surface

Derived Metrics

- Cone maximum height (from relief)
- Cone volume (positive relief × pixel area)
- Crater depth (rim percentile – floor elevation)
- Crater fill volume

Outputs: structured csv (60 metrics per vent), warning flags, and optional diagnostic messages and figures.

Stage 3: Automated Pipeline

Objective

Execute Stage I (segmentation) and Stage II (morphometrics) at scale using a two-phase parallel pipeline with resume support and fault tolerance.

Inputs: list of vent coordinates, polygon + DEM directories, output files for metrics, persistent failure log, and run log, configurable retry + backoff parameters, and functions from stages 1 and 2.

Phase 1 — Initial Parallel Processing

- Processes all new cones in parallel via ProcessPoolExecutor
- Safe concurrent writes to metrics CSV using a shared lock
- Errors classified as fatal or transient
- Global download protection prevents API overload, triggering cooldowns on repeated DownloadErrors

Error Classification:

Fatal (no retry)

- NullError
- DiskSpaceError
- CRS_Error
- BasalSurfaceError
- DEMSizeError

Transient (retryable)

- DownloadError
- Timeout / TimeoutError
- RasterioIOError
- Worker crash
- Unclassified runtime errors

Phase 2 — Targeted Retry

- Retries only transient failures with exponential backoff
- Per-cone retry limit enforced (MAX_TOTAL_ATTEMPTS)
- Cones escalated to FAILED_FATAL if retries are exhausted
- Retry loop continues until all cones succeed or an external outage triggers abort

Persistant Logging

- Run log tracks all processing events, retries, and errors
- Failure CSV preserves the cone state for automatic resume

Outputs: segmented DEMs, cone and crater polygons, csv of cone metrics, and failure and run logs.

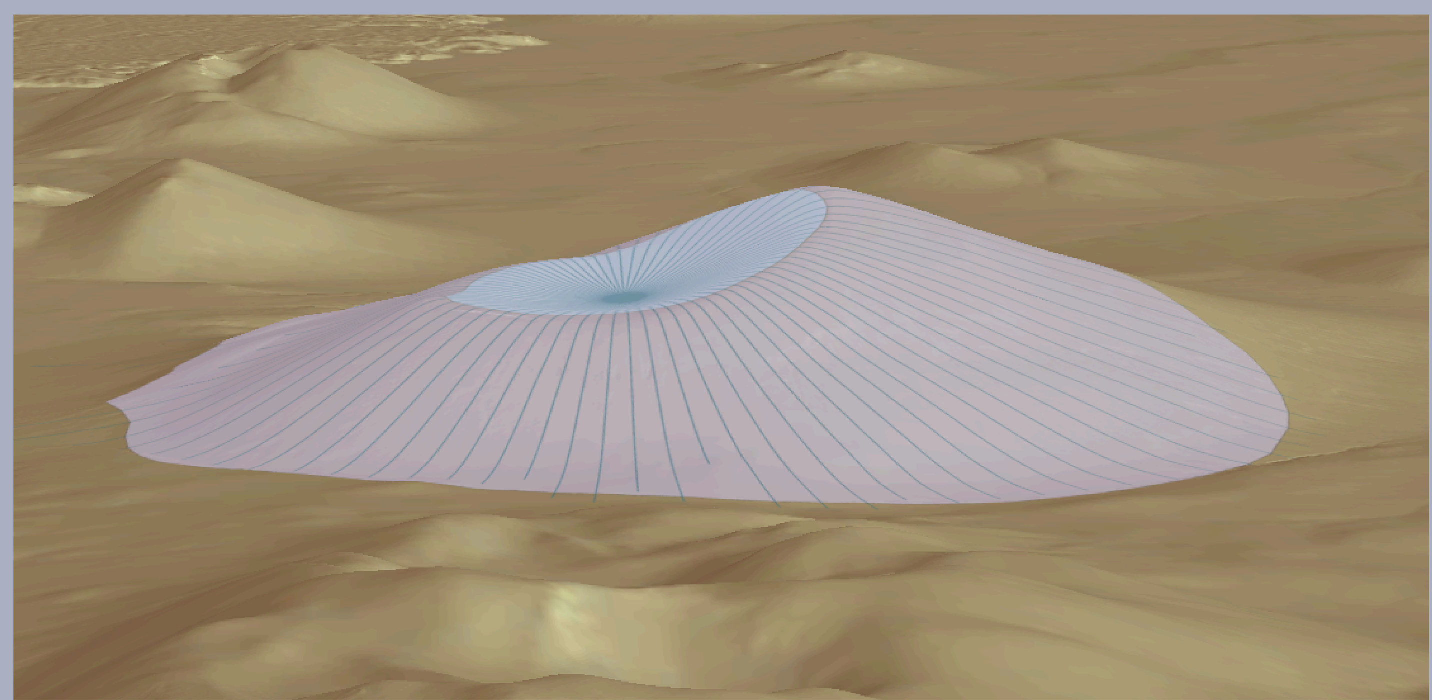


Figure 5: Cone 7, located at 35° 33' 30.83", -111° 36' 20.15"

References

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[2] Poulos M. J. et al. (2012), Geophys. Res. Lett. 39, 2012GL051283.
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[4] Brož, P.et al. (2015), JGR Planets 120, 1512–1527.
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[6] Mills A. et al. (2024), Icarus 418, 116145.